

GEOG 5103

Geographic Information Science: Spatial Analytics

FORMAT	CATEGORY	CREDITS
On-Campus	Elective	4
SECTION	BEST ROLE MATCH	AVG ALIGNMENT
Section 4 – Elective (On-Campus)	Machine Learning Engineer (60.2 / 100)	47.0 / 100

47

CURRENT AVG SCORE

across 6 job roles

85

POTENTIAL SCORE

with recommended additions

+56

MAX SCORE GAIN

best single-role improvement

CURRENT COURSE DESCRIPTION

Explores advanced topics in geospatial databases, spatial analytics and geoprocessing in a Geographic Information System (GIS). Emphasizes how geographic concepts are linked to methodological frameworks for recording, transforming, storing/retrieving, analyzing, and processing geographic data as well as various forms of uncertainty. Exercises demonstrate the application of GIS-based methods to real world scenarios in interdisciplinary settings.

INDUSTRY ALIGNMENT BY JOB ROLE

JOB ROLE	CURRENT SCORE	PROJECTED SCORE	POINT GAIN	PRIORITY
Machine Learning Engineer	60.2	99.7	+39.5	HIGH
ML Engineer	53.9	82.8	+29.0	MEDIUM
AI Engineer	53.0	86.9	+33.9	HIGH
Applied Scientist	43.9	100.0	+56.1	HIGH
Data Scientist	35.8	74.4	+38.7	HIGH
MLOps Engineer	35.3	65.6	+30.3	HIGH

RECOMMENDED TOPICS TO ADD

Keywords highlighted in gold (example) appear as gaps across 3 or more job roles.

System Design 3 of 6 roles

systems

LLMs & Generative AI 1 of 6 roles

storage

Deep Learning 1 of 6 roles

training

Business Communication 1 of 6 roles

business

PRIORITY RECOMMENDATIONS FOR INSTRUCTORS

1 **HIGH** **Add System Design coverage** — required vocabulary in 3 of 6 target roles. Integrate through lectures, assignments, or lab exercises. Key terms: systems.

2 **LOW** **Optional: LLMs & Generative AI** — would broaden appeal to one additional role. Candidate terms: storage.

3 **LOW** **Optional: Deep Learning** — would broaden appeal to one additional role. Candidate terms: training.

4 **LOW** **Optional: Business Communication** — would broaden appeal to one additional role. Candidate terms: business.

GAP KEYWORDS BY ROLE

Machine Learning Engineer

60.2 → 99.7 (+39.5 pts)

sets learning tools solutions business mobile design development

ML Engineer

53.9 → 82.8 (+29.0 pts)

media discovery learning machine systems models model problems

AI Engineer

53.0 → 86.9 (+33.9 pts)

following models learning software model systems problems technology

Applied Scientist

43.9 → 100.0 (+56.1 pts)

biology discovery scientific computational research storage problems models

Data Scientist

35.8 → 74.4 (+38.7 pts)

sets different social problems approaches

MLOps Engineer

35.3 → 65.6 (+30.3 pts)

model systems learning machine media training tools models

Methodology: Scores (0–100) reflect TF-IDF cosine similarity between course catalog descriptions and a corpus of 100 LinkedIn data science job postings (collected June 2026, filtered to postings with ≥ 200 applicants). Gap keywords are terms that appear frequently in job postings for a given role but are absent or underrepresented in the current course description. "Projected score" shows the estimated alignment after integrating the top 5 suggested keywords. Analysis was conducted by the CU Boulder MSDS Curriculum-Industry Alignment project team.